

The Universe's Beginning: A History of Contending Cosmologies from the Steady State to the Hubble Tension

The Genesis of Competing Cosmologies: Philosophical and Historical Context

The mid-twentieth century witnessed a profound shift in humanity's understanding of the cosmos, marked by a direct conflict between two fundamentally different models of universal existence: the Steady State theory and the Big Bang theory. This scientific dispute was not merely a technical disagreement over astronomical data; it was deeply embedded in the philosophical, methodological, and even theological currents of the era. To comprehend the full significance of this debate, one must first examine the intellectual landscape that preceded it, including the epistemic attitudes of key figures, the lingering influence of classical physics, and the philosophical repugnance many felt toward a universe with a singular beginning. The very concept of a temporal origin for the cosmos, or *creatio ex nihilo*, entered the realm of natural science only with the advent of modern cosmology in the mid-twentieth century [158](#). Prior to this, most physicists, including Albert Einstein, were inclined to believe the universe had always existed [83](#).

The groundwork for relativistic cosmology was laid by Einstein with his general theory of relativity [153](#). Initially, he sought a static model of the universe, leading him to introduce the cosmological constant to counteract gravitational collapse [153](#). However, the discovery of galactic recession by Edwin Hubble pointed towards an expanding universe, challenging this static picture [66](#). This led to the development of dynamic models, but the question of the universe's ultimate origin remained contentious. Prominent physicist Arthur Eddington, for instance, found the idea of a universe's beginning philosophically repugnant, citing both philosophical and observational concerns [61](#). His opposition to singularities and beginnings reflects a broader preference for timeless, unchanging models that echo ancient thought, such as Aristotle's changeless and eternal cosmos [61](#) [72](#). Similarly, the German physicist Walther Nernst argued that denying the infinite duration of time would be akin to abandoning the laws of nature [109](#). This philosophical

predisposition towards an eternal universe created fertile ground for alternatives to a model that inherently required a starting point.

This philosophical context heavily influenced the development of the Steady State theory. Its proponents, Hermann Bondi, Thomas Gold, and Fred Hoyle, explicitly built their model on the "Perfect Cosmological Principle," which extended the standard cosmological principle of homogeneity and isotropy to include time ^{31 55}. It stated that the observable universe is statistically unchanged over time ⁸⁸. For an expanding universe to appear unchanged, Bondi, Gold, and Hoyle proposed that matter must be continuously created to fill the voids left by expansion ¹⁶⁹. This continuous creation was not an ad hoc addition but a fundamental law of nature, incorporated into the basic laws of physics ¹⁰⁸. The motivation was partly pragmatic, as Bondi and Gold argued that without this principle, "cosmology" they wrote, "is no longer a science" ¹⁵⁴. However, it was also driven by a powerful philosophical desire to avoid a singular origin. For the atheistic Fred Hoyle, the idea of a universe beginning at a specific moment was philosophically unacceptable; he preferred a universe existing eternally in a Steady State ¹⁰⁷. This sentiment reveals a crucial aspect of the debate: the choice between competing cosmologies was often guided by deep-seated epistemic attitudes about what constitutes a satisfactory scientific explanation ¹⁰¹¹⁰².

In parallel, the Big Bang theory, though it implies a beginning, also emerged from a complex interplay of theoretical necessity and philosophical inquiry. Its origins can be traced to Georges Lemaître, a Belgian priest and physicist who, in 1927, proposed the "primeval atom" hypothesis ^{33 84}. He suggested the universe began as a single, giant atom that exploded and continues to expand ⁹⁴. While Lemaître himself saw no inherent conflict between his scientific work and his faith, viewing science as a unique human endeavor aimed at understanding God's creation, he personally opposed Pope Pius XII's attempts to use the theory as empirical validation for Christian theology of creation ^{30 67}. This highlights the sensitive intersection of cosmology and theology. Theologians like Thomas Aquinas had long distinguished between the "origin" of the universe and its "beginning," with beginning referring to a temporal start within an already existing framework of space and time ⁸². The Kalam cosmological argument, which posits that if something has a finite past, it must have a cause, further demonstrates how the Big Bang's implication of a beginning sparked intense philosophical and theological debate ¹⁰⁶¹²³. Thus, the stage was set for a great debate where two seemingly viable scientific models competed, each resonating with different philosophical intuitions about eternity, causality, and the nature of physical law.

The Steady State Theory: Premises, Proponents, and Claimed Evidence

The Steady State theory, formally articulated in 1948 by Hermann Bondi, Thomas Gold, and Fred Hoyle, presented a radical alternative to the prevailing notion of a universe with a beginning [21 121](#). Its foundation was the "Perfect Cosmological Principle," which postulated that the universe is not only homogeneous and isotropic in space but also completely independent of time [31 55](#). In essence, the universe as a whole appears the same to any observer at any point in time [88](#). To reconcile this eternal, unchanging appearance with the observed expansion of the universe described by Hubble's Law, the theory necessitated a continuous creation of matter [46](#). This process involved the spontaneous generation of new matter, primarily protons or hydrogen atoms, at an extremely low rate estimated to be roughly one particle per cubic meter per billion years [46 108](#). This continuous creation was considered a fundamental law of nature, integrated directly into the theory's framework [108](#). Hoyle, in particular, emphasized that his original formulation of the theory adhered closely to the framework of general relativity, distinguishing it from more kinematical approaches proposed by contemporaries like Milne [9](#).

The motivations for developing this theory were a blend of philosophical conviction and perceived scientific inadequacies of rival models. As previously noted, many scientists found the concept of a singular cosmic beginning philosophically repugnant [61 109](#). For Hoyle, an atheist, the idea of a "big bang" was anathema; he preferred a universe that was truly eternal, avoiding any need for a creator or a metaphysical singularity [107](#). This philosophical stance provided a strong impetus for constructing a model that could accommodate an expanding universe without requiring a temporal origin. Furthermore, the proponents believed their principle was essential for maintaining cosmology as a rigorous science, arguing that without it, the field would lose its predictive power and become speculative [154](#). The intellectual lineage of the theory also includes earlier, abandoned attempts. Albert Einstein himself drafted an unpublished manuscript around early 1931 attempting to construct a steady-state model, demonstrating that the idea predated the 1948 formulation by Bondi, Gold, and Hoyle [20 121162](#).

Proponents of the Steady State theory marshaled significant evidence, primarily from the burgeoning field of radio astronomy. The group at the Cavendish Laboratory in Cambridge, led by Martin Ryle, conducted extensive surveys of radio sources in the sky [53](#). They discovered that the number of faint radio sources was lower than expected in a simple, uniformly expanding universe model [52](#). Ryle and his colleagues interpreted this

deficit as evidence that the universe's properties do not change over time [135](#). If the universe were evolving, one would expect to see more distant (and therefore older) and more numerous faint sources. The apparent maximum in the number count distribution suggested a "steady-state" picture where the overall density of visible objects remains constant [166](#). Hoyle himself argued that these source counts were in agreement with the predictions of the Steady State theory, pointing to a minor deficiency of only five sources per steradian as a minor discrepancy rather than a fatal flaw [76](#). This body of observational data formed the primary empirical pillar supporting the Steady State model during its heyday. The theory's main competitor, the Big Bang, was seen by some as overly reliant on unproven assumptions, while the Steady State offered a clear, albeit extraordinary, mechanism—continuous creation—to explain the universe's large-scale properties.

Feature	Steady State Theory	Big Bang Theory
Core Principle	Perfect Cosmological Principle: Universe is unchanging in space and time 31 88 .	Universe had a hot, dense beginning approximately 13.8 billion years ago and has been expanding and cooling ever since 81 94 .
Key Authors	Hermann Bondi, Thomas Gold, Fred Hoyle (1948) 21 121 .	Georges Lemaître (1927), George Gamow, Ralph Alpher, Robert Herman (late 1940s) 12 33 .
Primary Motivation	Philosophical aversion to a singular beginning; desire for an eternal, unchanging cosmos 61 107 .	Theoretical explanation for Hubble expansion and light element abundances; prediction of remnant radiation 26 66 .
Mechanism for Expansion	Continuous creation of matter at a low rate (~ 1 proton/m ³ /yr) to maintain constant density 46 108 .	Intrinsic expansion of spacetime itself from an initial singularity 94 .
Claimed Evidence	Radio source counts showing a maximum, suggesting no evolution over time 52 166 .	Cosmic Microwave Background radiation, primordial nucleosynthesis element abundances, Hubble expansion 8 26 .

The Big Bang Paradigm: Development, Key Figures, and Empirical Corroboration

The Big Bang theory, now the cornerstone of modern cosmology, evolved gradually from a few visionary ideas into a comprehensive physical model. Its conceptual roots trace back to the 1920s. In 1927, the Belgian physicist and priest Georges Lemaître published a seminal paper in which he derived Hubble's Law and proposed that the universe originated from a "primeval atom" [33](#) [84](#). This "Fireworks Universe" became known as the Big Bang theory, for which Lemaître is widely regarded as the father [29](#) [81](#). However, it was not until after World War II that the theory was developed into a robust physical framework by George Gamow and his collaborators [12](#). Gamow, a Russian-born

physicist, became a pivotal figure in championing and developing the theory in the United States [65](#). Working with his students Ralph Alpher and Robert Herman, Gamow formulated a detailed model of the early universe [159](#). Their primary motivation was to solve the problem of the origin of the chemical elements. They proposed that all the elements were forged in the intense heat and pressure of the primeval fireball through a process they termed primordial nucleosynthesis [26](#) [161](#).

A critical and far-reaching contribution of Alpher and Herman was their prediction of a residual thermal radiation field left over from this hot, dense beginning [10](#) [163](#). In 1948 and 1949, they calculated that this "fireball" should now appear as a uniform background of microwave radiation with a temperature of around 5 K [100](#)[115](#). This prediction, however, largely went unnoticed by the mainstream physics community at the time [10](#). The theory gained significant traction when it was realized that its success in explaining the observed abundances of light elements (hydrogen, helium, deuterium, and lithium) was a powerful piece of evidence in its favor [26](#) [28](#). The precise match between the predicted and observed ratios of these light elements provided strong support for the high-density, high-temperature conditions posited by the Big Bang model.

The definitive corroboration of the Big Bang theory came from an unexpected source. In 1965, Arno Penzias and Robert Wilson, working at Bell Telephone Laboratories, detected a persistent, uniform microwave noise coming from all directions in the sky [23](#) [78](#). Initially baffled by this "excess antenna temperature," they consulted theorists, including Robert Dicke at Princeton, who immediately recognized the significance of the discovery [80](#). Dicke and his team had been actively searching for the very radiation that Penzias and Wilson had serendipitously found [63](#). The discovery of the Cosmic Microwave Background (CMB) was a turning point in cosmology [63](#). The CMB was precisely the kind of remnant radiation predicted by Alpher and Herman a decade earlier [10](#). Its measured blackbody spectrum and temperature of 2.7 K matched the Big Bang prediction with remarkable precision [93](#) [146](#). The discovery essentially discredited the Steady State theory, which had no mechanism to produce such a pervasive, thermal background radiation [8](#) [17](#). The CMB provided strong support for the hot Big Bang model, transforming it from a compelling hypothesis into the dominant scientific paradigm [114](#). Today, the Big Bang model has matured into the Lambda-Cold Dark Matter (Λ CDM) model, which incorporates dark energy (the cosmological constant, Λ) and cold dark matter to explain the universe's large-scale structure and accelerated expansion [96](#). This standard model is supported by a wealth of evidence, including the detailed anisotropies in the CMB, large-scale structure surveys, and observations of distant supernovae [77](#) [144](#).

Alternative Frameworks: Oscillating Universes and Plasma Cosmology

While the debate raged between the Steady State and Big Bang models, other cosmological frameworks were also being explored, motivated by different philosophical and theoretical considerations. Among the most prominent were oscillating (or cyclic) universe models and plasma cosmology. These alternatives sought to address perceived shortcomings in the dominant paradigms, offering visions of the cosmos that differed significantly in their fundamental mechanisms and historical scope.

Oscillating universe models propose a cyclical cosmos that undergoes endless sequences of expansion and contraction [148](#). In this view, the Big Bang is not a singular beginning but merely the start of the current expansionary phase of an eternal series of cycles [86](#) [105](#). Each cycle begins with a "bang" and ends with a "crunch," followed by another big bang [152](#). This concept is one of the earliest relativistic cosmological models, first introduced by Alexander Friedmann in his seminal 1922 papers [40](#) [71](#). Later, Richard Tolman developed a more detailed version of the oscillating universe model [41](#) [42](#). The primary motivation for such models was to circumvent the initial singularity problem inherent in the standard Big Bang theory—the breakdown of physical laws at a point of infinite density and temperature. By envisioning an eternal succession of universes, oscillating models offered a way to describe cosmic evolution without a true beginning [149](#). Some versions of the model incorporate mechanisms to manage entropy across cycles, such as an early period of acceleration that prevents entropy from diverging [42](#). More recent incarnations of cyclic models have been inspired by string theory, notably the model proposed by Paul Steinhardt and Neil Turok, which replaces the singularity with a smooth transition between cycles [44](#). Despite their theoretical appeal, oscillating models face significant challenges, including the difficulty of explaining how a universe can transition smoothly from a state of maximum compression to one of maximum expansion and how information from a previous cycle would not affect the new one.

Another significant alternative is plasma cosmology, a theory championed by Nobel Prize-winning physicist Hannes Alfvén [6](#) [50](#). This model posits that electromagnetic forces, acting on the vast networks of electrically charged particles (plasmas) permeating the universe, are the dominant force shaping cosmic structures, not gravity [49](#). Alfvén and his followers argue that many astrophysical phenomena, such as the formation of galaxies and the emission of radiation from quasars, can be explained by plasma processes without recourse to exotic concepts like dark matter, dark energy, or the hot Big Bang scenario [167](#). The theory draws on Alfvén's own pioneering work in

magnetohydrodynamics and his investigations into the occurrence of plasmas in space [50](#) . Proponents of plasma cosmology cite observational evidence, such as the filamentary structures seen in galaxy distributions and synchrotron radiation from cosmic jets, as manifestations of large-scale Birkeland currents—spiral-shaped electric currents in space [74](#) [91](#) . They contend that the standard Big Bang model has become overly reliant on unobserved entities to fit the data [140](#) . However, plasma cosmology remains a non-mainstream and highly controversial framework. It struggles to account for key successes of the Big Bang model, most notably the precise abundances of light elements from primordial nucleosynthesis and the detailed statistical properties of the Cosmic Microwave Background [4](#) [100](#) . While some researchers continue to advocate for its reconsideration, viewing it as a potentially fruitful alternative to the increasingly complex standard model [5](#) [47](#) , it lacks the broad empirical support and predictive power that established the Big Bang as the prevailing paradigm.

The Demise of Steady State: A Case Study in Falsifiability and Scientific Progress

The dramatic decline of the Steady State theory and the ascent of the Big Bang model serves as a canonical example in the philosophy of science, illustrating the principles of falsifiability and the dynamics of scientific research programmes. The conflict was not resolved by abstract arguments alone but by decisive empirical tests that pitted the specific, testable predictions of each theory against observational data. The story of the Cosmic Microwave Background (CMB) radiation is the central narrative of this scientific resolution, acting as a "crucial experiment" that ultimately settled the debate in favor of the Big Bang [17](#) .

From a perspective rooted in Karl Popper's philosophy of science, a theory's scientific status hinges on its falsifiability—the capacity for it to be proven wrong by observation or experiment [89](#) [111](#) . Both the Steady State and Big Bang theories were, in principle, falsifiable. The Steady State theory made several sharp predictions, most notably concerning the number counts of radio sources. It predicted that the number of sources brighter than a given flux should increase with distance up to a point, then decrease, forming a peak [166](#) . When the Cambridge group's 2C survey data showed a slope inconsistent with this prediction, it constituted a direct challenge to the theory [166](#) . Although proponents attempted to salvage the model by questioning the analysis and instrumentation of the surveys [51](#) , the growing body of conflicting data weakened its

empirical standing. More importantly, the Steady State theory contained a single adjustable parameter: the rate of matter creation [130](#). While this was seen by some as a virtue of simplicity, it also meant the theory lacked the predictive power of the Big Bang, which could make novel predictions based on established physics without free parameters.

The final blow to the Steady State theory came from its inability to account for the CMB. The discovery of this pervasive microwave radiation in 1965 was a complete surprise to the radio astronomers who found it [23](#) [63](#). However, theorists like Robert Dicke quickly realized its significance, interpreting it as the cooled remnant of a hot, dense early universe [80](#). The Big Bang model, with its prediction of this relic radiation by Alpher and Herman, was vindicated overnight [10](#) [163](#). The CMB's perfect blackbody spectrum and uniform temperature provided overwhelming evidence for a thermal origin consistent only with the Big Bang's fiery beginning [114](#)[146](#). The Steady State theory, lacking any mechanism for producing such a thermal bath of photons, was rendered untenable [8](#). The weight of this single piece of evidence was so great that it essentially ended the viability of the Steady State model as a serious contender in cosmology [17](#).

An analysis through the lens of Imre Lakatos's methodology of scientific research programmes offers a complementary perspective. According to Lakatos, science progresses through competing "research programmes," each composed of a "hard core" of central hypotheses and a "protective belt" of auxiliary hypotheses that can be modified to protect the hard core from falsification. A progressive programme generates novel predictions that are confirmed, while a degenerative one becomes increasingly ad hoc in its attempts to explain away contradictory evidence. Viewed through this framework, the Big Bang programme was clearly progressive. Its hard core—that the universe evolved from a hot, dense state—was continually bolstered by novel, spectacularly successful predictions. The prediction and discovery of the CMB was the most important of these. In contrast, the Steady State programme, while initially progressive with its radio source count predictions, soon became degenerative. As more and more data conflicted with its core tenets, its proponents resorted to modifying auxiliary assumptions (e.g., about instrumental errors) rather than revising the theory's fundamental premise of continuous creation [51](#). The failure of the CMB prediction to be replicated in the Steady State framework marked the definitive moment when one research programme surpassed the other, leading to the near-universal acceptance of the Big Bang model.

Contemporary Crisis in Cosmology: The Hubble Tension and the Future of Λ CDM

The triumph of the Big Bang theory and its consolidation into the standard Λ CDM model ushered in an era of precision cosmology. However, the twenty-first century has revealed that this consensus may be fraying at the edges. A growing collection of high-precision observations has uncovered significant tensions between measurements of the universe's expansion rate, a discrepancy known as the "Hubble tension" [57](#) [99](#). This modern crisis echoes the great debate of the mid-twentieth century, suggesting that our current understanding of the universe's origin and composition may be incomplete.

The Hubble tension refers to the substantial difference between two values for the Hubble constant (H_0), the current rate of the universe's expansion [99](#). On one hand, "early-time" measurements infer H_0 by observing the Cosmic Microwave Background (CMB) from the Planck satellite and then extrapolating forward using the Λ CDM model. This yields a value of approximately 67 km/s/Mpc. On the other hand, "late-time" measurements determine H_0 by observing local astronomical objects like Type Ia supernovae and baryon acoustic oscillations (BAOs) [77](#). These direct measurements yield a higher value of approximately 73 km/s/Mpc [60](#). The discrepancy between these two values is now well-established and statistically significant, representing a major challenge to the Λ CDM paradigm [95](#) [96](#). Many researchers interpret this tension as a sign of "new physics" beyond the standard model, perhaps involving modifications to dark energy, dark matter, or even gravity itself [58](#) [141](#). Other related anomalies, such as the σ_8 tension concerning the clustering of matter and various anomalous features in the CMB, further compound the sense that Λ CDM may be an incomplete description of reality [97](#) [144](#).

This contemporary situation bears a striking resemblance to the historical conflict between the Steady State and Big Bang theories. Just as radio source counts in the 1950s and 60s provided the first major challenge to the simplicity of the Big Bang, the Hubble tension and other anomalies are now providing the most significant challenge to the elegance of the Λ CDM model. Then, a single, powerful piece of evidence—the CMB—resolved the debate. Now, a persistent discrepancy in measurement techniques threatens to open the door to new theoretical frameworks. Some researchers are exploring extensions to Λ CDM, such as models with early dark energy, decaying dark matter, or interacting dark sectors, in an attempt to resolve the tension within a familiar paradigm [129](#)[140](#). Others look toward more radical departures, reviving interest in alternative cosmologies that were once sidelined. Ideas reminiscent of cyclic models or plasma cosmology, for instance, might offer different explanations for the observed

discrepancies. The Hubble tension problem is one of the most significant challenges in modern cosmology, driving a search for solutions that might imply new physics [58](#). The resolution of this modern puzzle will likely shape the next chapter in our quest to understand the origin and fate of the universe, potentially heralding a new paradigm shift analogous to the one that brought the Big Bang to dominance over the Steady State.

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